Acute Toxicity Testing

Acute toxicity testing, required by many regulatory frameworks, evaluates the general hazardous effects of a chemical or product during a 21-day observation period (Walum 1998). It helps to classify and label the products for better risk management in chemical manufacturing, distribution and consumption. However, the methods commonly used for acute toxicity testing can be quite questionable and unethical as it is usually determined in vivo (performed on living organisms). The chemical toxicology classification is done via Lethal Dose 50 (LD_{50}) testing in which it seeks to identify the single dose of a substance that will kill 50% of the test group's animals. As such, there have been efforts to develop predictive *in silico* (produced by means of computer modelling or simulation) models for acute oral systemic toxicity through global partnerships. A variety of diverse models by researchers will be aggregated to overcome the shortcomings of individual approaches and thus allow for regulatory acceptance of computational predictions to replace animal testing (Kleinstreuer et al. 2018). The creation of such *in silico* models can be done using a substantial body of rat acute oral lethality data that was gathered by the NTP Interagency Center for the Evaluation of Alternative Toxicological Methods (NICEATM), in collaboration with the U.S. Environmental Protection Agency's National Center for Computational Toxicology (U.S. EPA NCCT).

Data Set

The data set consists of five different modelling endpoints to be used in predictions, for example very toxic and nontoxic endpoints. A study by Ballabio et al. (2019) has used QSAR models based on a Bayesian consensus approach that integrates three classification algorithms, namely N-Nearest Neighbours (N3), Binned-Nearest Neighbours (BNN) and Naïve Bayes (NB). However, for our simple analysis, we will be using only part of the data set used by Ballabio et al. (2019) to build a logistic regression model for predicting very toxic and not very toxic molecules. The Quantitative Structure-Activity Relationship (QSAR) oral toxicity data set is a multivariate data set with 8992 instances from two classes, 1024 attributes and an experimental class of positive and negative labels for classification purposes. The instances, from the very toxic endpoint, have ground truth labels of very toxic/positive (741 instances) and not very toxic/negative (8251 instances), where the data set is imbalanced since the positive class is very much less represented.

Here, instances are labelled as positive if their LD_{50} values are less than 50 mg/kg, and labelled negative otherwise. The 1024 attributes are essentially binary molecular fingerprints of 0's and 1's that encode the existence or absence of specific molecular fragments and substructures (Todeschini & Consonni 2009) for model calibration.

Logistic Regression Model

In this study, one of the most famous supervised machine learning algorithms, logistic regression, is utilised to build a model for classifying the oral toxicity data besides computing the test error rate of the classification model built. It is specifically a binomial logistic regression following the fact that there are only two possible types of dependent variables, namely positive or negative. The logistic regression model is suitable to predict a categorical dependent variable's output given a set of independent variables besides restricting probabilistic values of the output to be between 0 and 1. By default, probability scores greater than a threshold value of 0.5, will have the sample classed in the positive class (1), whereas samples with probability scores below 0.5 will be classed in the negative class (0). Logistic regression models are efficient to train, implement and interpret besides the fact that they can classify unknown entries quickly (Raj 2020). It can also interpret model coefficients as measures of the significance of an independent variable by observing the summary results obtained through the *statsmodels.formula.api* library. The logistic regression model obtained for this analysis can be represented by the following general formula:

$$y = \frac{e^{\beta_0+\beta_1 X_1+\beta_2 X_2+\cdots+\beta_{1024}X_{1024}}}{1 + e^{\beta_0+\beta_1 X_1+\beta_2 X_2+\cdots+\beta_{1024}X_{1024}}}$$

Let $t = e^{\beta_0+\beta_1 X_1+\beta_2 X_2+\cdots+\beta_{1024}X_{1024}}$. Then,

$$y = \frac{e^t}{1+e^t} = \frac{e^t}{1+\frac{1}{e^{-t}}} = \frac{e^t}{\frac{e^{-t}+1}{e^{-t}}} = \frac{e^{t-t}}{e^{-t}+1} = \frac{1}{1+e^{-t}}$$

Since the values of y will always be between 0 and 1, y can be set as the probability given some value x. Thus, the final general formula for our logistic regression model is:

$$\log\frac{p(X)}{1-p(X)} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_{1024}X_{1024}$$

where $\frac{p(X)}{1-p(X)}$ is called "odds", that is the probability of X happening over the probability that X does not happen. Here, log odds are modelled by a linear model, which can be solved by linear regression. For example, given a one unit increase in X_2, the log odds will increase by the value of β_2.

Procedure

The procedures for building the classification model are explained in this section. Firstly, we observe the data set read into Jupyter using the pandas library. It has 1024 columns of binary attributes, while the last column is the ground truth labels of "negative" and "positive". Since this machine learning technique cannot handle variables that are not in numeric format, the "negative" labels are set to the value 0 and the "positive" labels are set to the value 1 in a new column called "Positive". The earlier ground truth labels' column is then dropped from the data set. The binary attributes are then separated into the "X" variable, while the "Positive" column is saved in the "y" variable to enable the splitting of the data set into training and testing sets. In the study by Ballabio et al. (2019) the researchers used the 75:25 rules for the training and testing sets, where they also took into account the proportions of positive and negative molecules in both sets. However, we make some changes by utilising scikit-learn's *train_test_split* function, using 80% of the data for training and the other 20% for testing. The random state is set to 25 in this study. Here, only molecules in the training set are used in parameter adjustments to optimise the models, whereas the molecules in the testing set examines the model's ability to adapt to new and unseen data. After fitting the logistic regression model on the training data, metrics to assess the performance of the model can be computed on the testing data. The performance of the model need not be assessed on training data as there are high chances to overestimate the model's capabilities since the training data has already been memorised by the model. The testing data will be able to provide a more accurate estimation of the model's performance, which might be fine or need improvements if the results are unsatisfactory.

Results and Discussion

To access the performance of our model fitting, the confusion matrix, accuracy score and test error rate for the logistic regression model are listed in the table below and discussed.

Performance Metric	Output from Jupyter
Confusion Matrix	`array([[1624, 35], [ 75, 65]], dtype=int64)`
Accuracy Score	0.9388549193996665
Test Error Rate	0.06114508060033352

From the table above, we see that out of the 1799 molecules, 1624 (True Negatives, TN) and 65 (True Positives, TP) have been correctly classified by the model. These are instances where the model has correctly predicted the negative and positive classes respectively. This leaves us with 110 molecules that were wrongly classified. 75 are False Negatives (FN), meaning that the model predicted it as a negative when it was actually a positive. 35 are False Positives (FP), where the model predicts them as positives when they are actually negatives. Besides using the *accuracy_score* function in scikit-learn, the accuracy score of 93.89% can be obtained by dividing the number of correct predictions over the total number of predictions as follows:

$$Accuracy\ Score = \frac{TP + TN}{TP + TN + FP + FN} = \frac{65 + 1624}{65 + 1624 + 35 + 75} = 0.9389$$

On the other hand, the test error rate is simply the proportion of wrongly classified molecules in the test set or in other words:

$$Test\ Error\ Rate = 1 - Accuracy\ Score = 1 - 0.9389 = 0.0611$$

The test error rate is actually an important metric to evaluate a model's performance since higher test error rates will simply indicate that the model is overfitting to the training data and unable to generalise to newer, unseen data. Reducing test error rates should be of significant importance as it ensures more accurate predictions. Here, the logistic regression model has achieved a test error rate of 6.11%.

Let's take a closer look at positive and negative classifications to gain more insights as to whether 93.89% is really a good accuracy score in determining very toxic molecules or not. Out of the 1659 negative values, 1624 were correctly predicted, leading to a 97.89% accuracy for negative values. However, out of the 140 positive values, only 65 were correctly identified, that is a 46.43% accuracy for positive values. This actually means that 75 very toxic molecules go unaccounted for. A 93.89% accuracy might have looked good at a glance before we realised that the model did not do such a good job in predicting positive values. Therefore, this indicates that accuracy and test error rate does not paint the whole picture when a class-imbalanced data set is involved as it may only correctly predict the majority class without capturing the minority class. Other metrics such as recall or precision could be analysed further for class-imbalanced data sets like these.

Unsuitability of LDA and QDA for This Data Set

In this section, the unsuitability of the Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) for the QSAR oral toxicity data set is discussed. LDA and QDA are not suitable for this data set mainly because these two techniques assume that all of their predictor variables come from a gaussian distribution, or in other terms assumes that the independent variables are normally distributed. LDA also assumes equal covariances between classes, whereas QDA allows for non-equal covariances between classes. A normal distribution is essentially a continuous distribution that has a bell-shaped curve and is symmetric about the mean. Since the data set takes on only two values, it is actually a categorical data, not continuous data and cannot be modelled by the normal distribution. Besides that, the data is highly skewed, seeing as there is much more negative instances than there are positive instances. The high skew and imbalance in the data set favouring towards the not very toxic class, further emphasises that it is not from a normal distribution and violates the normality assumption. If LDA or QDA were to be used on the data set, poor classification results might be obtained due to their inability to represent the underlying data distribution. It is also worth noting that LDA and QDA are much more well suited for multiclass classification problems.

All in all, a logistic regression model is more appropriate for the QSAR oral toxicity data set since it can handle the categorical predictor variables and is more well-known for binary classification problems. Besides that, logistic regression does not require the data to come from a normal distribution and does not have equal or unequal covariance assumptions across classes.

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